

Article title

SmartSense Monitoring: an open-source wireless PT100 node and gateway for long-term surface-temperature monitoring of industrial equipment

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Abstract

SmartSense Monitoring is an open-source wireless system for measuring and logging the surface temperature of industrial equipment. Each battery-powered node combines a Seeed Studio XIAO ESP32-C6 microcontroller, a three-wire PT100 probe, a MAX31865 RTD front-end and an Ebyte E220 LoRa radio at 433 MHz. A gateway based on a second ESP32-C6 with an integrated touch display receives the frames, stores a per-sensor daily history on a microSD card (with LittleFS as a fallback) and serves a local web dashboard over Wi-Fi, requiring no cloud service or permanent Internet connection. Unlike a handheld spot measurement, the system preserves the continuous 24-hour thermal behaviour of each machine, enabling the detection of trends and anomalies. Over a 32-day pilot deployment with three nodes (51,338 samples), the LoRa link availability exceeded 97% once scheduled plant stoppages were excluded, and the continuous history revealed a temperature rise on a suction-fan motor later associated with a bearing anomaly. All design files, firmware and the bill of materials are released under open-source licenses.

Keywords

Open-source hardware; LoRa; PT100; RTD; MAX31865; ESP32-C6; industrial temperature monitoring; predictive maintenance

Specifications table

Hardware name	SmartSense Monitoring
Subject area	• Engineering and material science • Educational tools and open source alternatives to existing infrastructure
Hardware type	• Measuring physical properties and in-lab sensors • Field measurements and sensors • Electrical engineering and computer science
Closest commercial analog	Industrial wireless RTD/condition-monitoring loggers such as the Advantech WISE-2410 and commercial PT100 wireless data loggers.
Open source license	Hardware: CERN-OHL-S v2. Firmware and software: MIT License. Documentation: CC-BY 4.0.
Cost of hardware	Approximately 170 USD for three nodes and one gateway (see Bill of materials). Approximately 35 USD per additional node.
Source file repository	Available in the article (to be deposited at Zenodo with a DOI upon acceptance).

1. Hardware in context

Continuous temperature monitoring of motors and other industrial equipment helps identify conditions associated with friction, overload, poor ventilation or mechanical degradation. In a manual inspection, an operator obtains only the value at the instant of measurement; if that value is not systematically recorded, the thermal evolution of the equipment cannot be reconstructed or related to its operating and stop cycles.

Wireless sensor networks distribute measurement points without installing communication cabling at each location [1,2,6]. For low-data-rate applications, LoRa provides a long-range, low-power alternative suited to battery-powered nodes, whose range and scalability have been characterised experimentally [3,7,8]. LoRa has been used for temperature and condition monitoring in industrial and infrastructure settings [9,10], within the predictive-maintenance paradigm of Industry 4.0 [4,11]. However, published open-source LoRa nodes typically target ambient or DHT-class humidity/temperature sensors and rely on a cloud back-end. SmartSense Monitoring differs in three respects relevant to industrial surface monitoring: (i) it uses an industrial-grade three-wire PT100 with a MAX31865 front-end rather than a low-cost ambient sensor; (ii) it stores a per-sensor daily history locally and serves the dashboard from the gateway itself, so it operates without Internet; and (iii) the design targets replication and customization, releasing the schematics, firmware and bill of materials under open-source licenses.

A PT100 RTD was chosen over a digital sensor such as the DS18B20 for several reasons: the PT100 is the industrial standard (IEC 60751) with excellent long-term stability and low drift; its resistive nature allows precise analog conditioning through the MAX31865, a purpose-built RTD-to-digital converter for two-, three- and four-wire PT100 with lead-resistance compensation; it offers a far wider range (-200 to 850 °C) than the DS18B20 (-55 to 125 °C), suitable for industrial surfaces that may exceed the digital sensor's limit; and it is available in surface-mount and magnetic probe forms appropriate for mounting on motors.

Compared with proprietary industrial loggers such as the Advantech WISE-2410, SmartSense Monitoring trades integrated vibration sensing for a lower cost, full openness and local-first operation, making it suitable for pilot plants and educational settings where flexibility and reproducibility matter more than a turnkey platform.

2. Hardware description

SmartSense Monitoring consists of one or more battery-powered sensor nodes and a gateway/visualizer. Each node acquires the surface temperature of a machine with a magnetic PT100 probe and a MAX31865 RTD front-end, estimates its battery state, and transmits the reading over a 433 MHz LoRa link before entering deep sleep. The gateway receives the frames, identifies each node by its MAC address, applies a configurable offset, stores the samples locally and presents them on an integrated touch display and a Wi-Fi web dashboard.

- It provides continuous, unattended surface-temperature logging of industrial equipment at a fraction of the cost of commercial wireless RTD loggers, using widely available modules.
- It operates fully offline: the gateway serves its own dashboard over a local Wi-Fi access point, so no cloud account or Internet connection is required.
- It preserves a per-sensor daily history, enabling the study of thermal trends, transients and machine on/off cycles that a spot measurement cannot capture.
- It is a modular basis for further work: additional nodes, alternative RTD probes, or extra variables (e.g. current, vibration) can be integrated by reusing the node design.
- It is suitable as a teaching platform for wireless sensor networks, RTD signal conditioning, embedded firmware and edge data handling, since every layer is open and documented.

Design files

The complete design files (KiCad schematics, firmware, bill of materials) are provided with the article and will be deposited in an open repository (Zenodo) with a DOI upon acceptance. The schematics are provided in editable KiCad format; the firmware source is fully included and commented.

3. Design files summary

Design filename	File type	Open source license	Location of the file
Node schematic (nodo_sensor)	KiCad schematic (.kicad_sch) + PDF/SVG	CERN-OHL-S v2	Available with the article
Gateway schematic (gateway)	KiCad schematic (.kicad_sch) + PDF/SVG	CERN-OHL-S v2	Available with the article
smartsense.kicad_sym	KiCad symbol library	CERN-OHL-S v2	Available with the article
Node firmware (Nodo_Lora)	Arduino C++ (.ino)	MIT	Available with the article
Gateway firmware (maestro_Lora_pantalla)	Arduino C++ (.ino) + web assets	MIT	Available with the article
Bill of materials (BOM.csv)	CSV spreadsheet	CC-BY 4.0	Available with the article
Enclosure (node/gateway)	3D model (STL + source CAD)	CERN-OHL-S v2	To be added

The node and gateway schematics document every external connection (SPI to the MAX31865, UART to the E220, switched sensor power, and the battery divider), verified against the firmware. The firmware files contain the node acquisition/transmission loop and the gateway reception, storage and dashboard logic. The bill of materials lists every purchasable component. The enclosure design files (PLA, 3D-printed) will be added to the repository.

Bill of materials

Costs are given in Colombian pesos (COP) with an approximate US dollar (USD) equivalent, based on the actual procurement prices of the project. The table below covers three nodes and one gateway. Prices vary by supplier and availability.

4. Bill of materials summary

Designator	Component	Number	Cost per unit (COP)	Total (COP)	Source of materials	Material type
U1	Seeed Studio XIAO ESP32-C6 (node)	3	30,000	90,000	seeedstudio.com	Semiconductor
U2	ESP32-C6 Touch LCD 1.47 board (gateway)	1	90,000	90,000	Local distributor	Semiconductor
U3	Ebyte E220-433 LoRa module + antenna	4	30,000	120,000	ebyteiot.com	Semiconductor
RT1	Magnetic three-wire PT100 probe	3	30,000	90,000	Local distributor	Metal
U4	MAX31865 RTD front-end module	3	14,000	42,000	sigmaelectronics.com	Semiconductor
BT1	LiPo battery 3.7 V 2000 mAh	3	30,000	90,000	didacticaselectronics.com	Other
U5	LiPo charge/protection module (TP4056 or equiv.)	3	5,000	15,000	Local distributor	Semiconductor
R1	Resistor 344.8 k Ω (battery divider)	3	—	—	Local distributor	Other
R2	Resistor 994.3 k Ω (battery divider)	3	—	—	Local distributor	Other
MEC1	3D-printed PLA enclosure	4	10,000	40,000	Self-printed	Polymer
W1	Wires, connectors, headers, hardware, adhesives	1	50,000	50,000	Local distributor	Other

Approximate total: 627,000 COP ($\approx$ 155 USD) plus a contingency allowance. The battery-divider resistors are drawn from a general parts kit and are accounted for in the wiring line item.

5. Build instructions

The following steps describe the construction of one sensor node and the gateway. Refer to the node and gateway schematics (Design files) and to the Bill of materials for component designators.

Sensor node

1. Solder the MAX31865 module in three-wire mode (bridge the 2/3-wire jumper), and connect the PT100 leads: the two leads from the same probe end to F+ and RTD+, and the remaining lead to RTD-, bridged to F-.
2. Wire the MAX31865 SPI bus to the XIAO ESP32-C6: CLK $\rightarrow$ D6 (GPIO16), SDO $\rightarrow$ D5 (GPIO23), SDI $\rightarrow$ D4 (GPIO22), CS $\rightarrow$ D3 (GPIO21). Power the module VIN from D2 (GPIO2) so the firmware can switch it off during deep sleep.
3. Connect the E220 LoRa module by UART: E220 TXD $\rightarrow$ D7 (GPIO17), RXD $\rightarrow$ D8 (GPIO19), M0 $\rightarrow$ D10 (GPIO18), M1 $\rightarrow$ D9 (GPIO20); power VCC/GND from 3V3/GND.
4. Build the battery divider: VBAT – R1 (344.8 k Ω) – A0 (GPIO0) – R2 (994.3 k Ω) – GND. Connect the LiPo cell through the TP4056 charge/protection module to the XIAO battery pads.
5. Flash the node firmware (`Nodo_Lora.ino`) with the Arduino IDE and the ESP32 board package. Install the Adafruit_MAX31865 library. Set RREF to the measured reference resistance of your MAX31865 module (421.1 Ω in this build).
6. Fit the components in the 3D-printed PLA enclosure and attach the PT100 magnetically to the target surface.

Gateway

1. Connect the E220 module to the ESP32-C6 Touch LCD board: E220 TXD $\rightarrow$ GPIO7, RXD $\rightarrow$ GPIO8, and both M0 and M1 to GPIO9; power VCC/GND from 3V3/GND.
2. Optionally wire a critical-alarm output (LED, buzzer or relay driver) to GPIO16.
3. Flash the gateway firmware (`maestro_Lora_pantalla.ino`) and upload the web dashboard assets to the on-board storage (LittleFS). Insert a microSD card for the history.

Safety. Use a fused LiPo cell with the TP4056 protection module and observe LiPo handling precautions. Ensure the PT100 and enclosure are rated for the surface temperature of the monitored equipment and mounted so they do not interfere with moving parts.

6. Operation instructions

1. Power the gateway. It creates a Wi-Fi access point (`Sensores_Temp`) and serves the dashboard at <http://192.168.4.1>.
2. Power each node. On first boot it transmits periodically (default configurable between 5 and 7200 s; 120 s in the pilot). The gateway registers each node by its MAC address.
3. In the dashboard, assign names to sensors, group them by machine, set the sampling interval, apply a per-node temperature offset, and configure warning and critical thresholds.
4. Monitor the live values (temperature, battery, connectivity) and consult the per-day 24-hour history, including minimum, maximum and average, and compare sensors of the same machine.



Figure 1: Assembled sensor node mounted on a motor housing: the PT100 probe with the MAX31865 module (left) and the node enclosure with the LoRa antenna (right).

5. To analyse data off-line or export figures, download the per-sensor history; the accompanying software replica can bulk-import the exported CSV.

Monitoreo de Temperatura Industrial

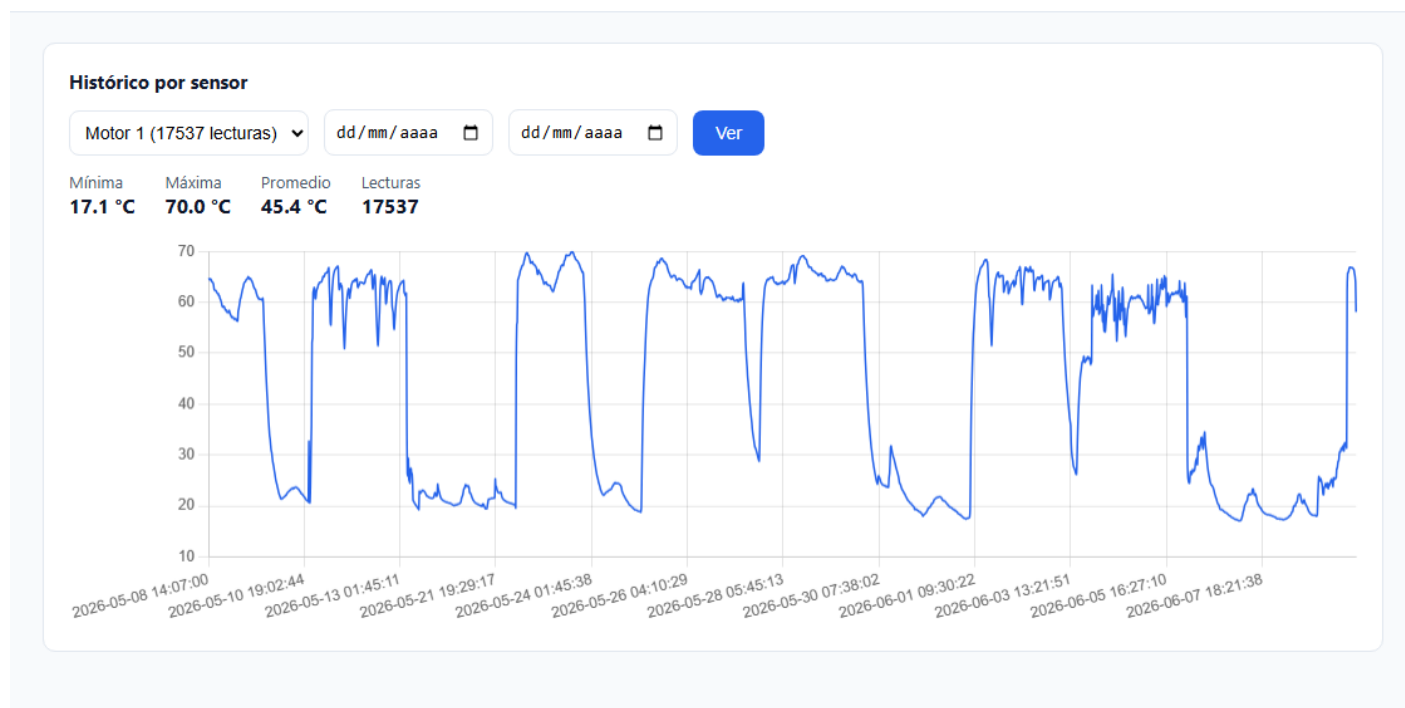


Figure 2: Local web dashboard: per-sensor temperature history over the deployment, with minimum, maximum and mean statistics. Shown here running on the open-source software replica with generic sensor labels.

Safety. The critical-alarm output is a supervisory aid, not a safety interlock; do not use it as the sole protection for equipment or personnel.

7. Validation and characterization

The system was validated in a pilot plant with three nodes over a 32-day deployment, producing 51,338 stored samples with a median sampling interval of 121 s.

Availability. Data outages were found to occur *simultaneously* across all three nodes, with identical start times (Fig. 6). This pattern indicates that the interruptions were periods with the gateway powered off (plant and power stoppages) rather than LoRa link failures, which would affect a single node at a time. Accordingly, Table 1 reports both the raw availability over the full deployment and the link availability computed over the active intervals only (i.e. excluding the documented gateway-off periods). The link availability exceeded 97% for all three nodes.

Node	Samples	Raw avail.	Link avail.	Stoppages	Hours off
Node A	17,537	76.0%	98.4%	66	176
Node B	16,487	32.3%	98.3%	73	1154
Node C	17,314	75.1%	97.8%	81	180

Table 1: Availability per node. Link availability excludes gateway-off periods, identified by their simultaneity across nodes. Node B includes a prolonged ~ 33 -day stoppage.

Reception cadence and packet loss. During active intervals the frame inter-arrival time was tightly concentrated around the 121 s median (Fig. 7), confirming a stable reception cadence. Estimating losses from the arrival intervals (a gap of k times the cycle implies $k - 1$ missing frames), the packet loss during active operation was 3.5–4.7% across the three nodes, i.e. a packet delivery ratio of 95–97%.

Representative 60-minute test. A representative one-hour window with the three nodes active (Fig. 3) illustrates a continuous test: at a 120 s cycle, 30 frames are expected per node in 60 min; 29, 30 and 30 were received (PDR 96.7%, 100%, 100%). Temperature and PDR are captured in this window; battery is characterised separately (Fig. 4) and recovery after a reset is 2.279 s. RSSI and SNR are not exposed by the current E220 firmware configuration, and end-to-end latency requires node–gateway clock synchronisation; both are declared as future work. The node frame now carries a sequence counter, enabling a fully instrumented sent-vs-received PDR in a repeated test.

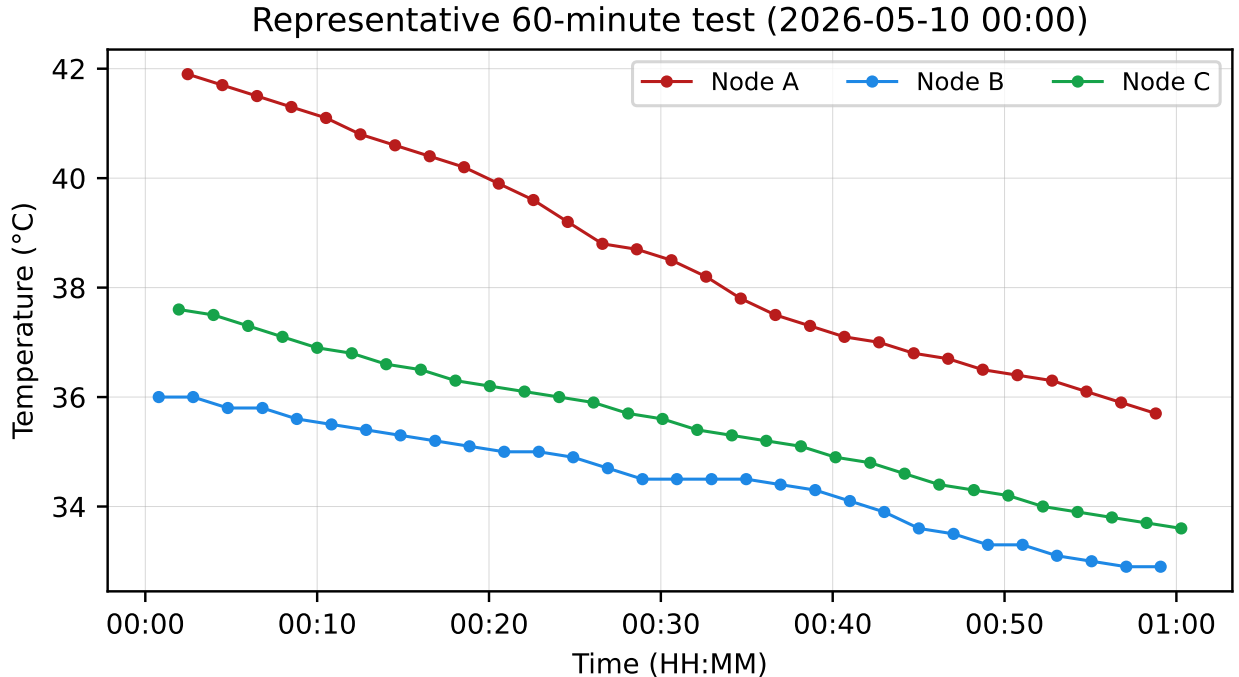


Figure 3: Representative 60-minute window (three nodes active) showing temperature versus time during a night-time cool-down of the machines.

Temperature agreement. Five paired readings were taken simultaneously at the same surface point of the suction-fan motor, in the 65–71 °C range, against a Fluke 62 MAX infrared thermometer (accuracy ± 1.5 °C or $\pm 1.5\%$, fixed 0.95 emissivity, used as a field reference without a current calibration certificate). The SmartSense readings agreed with the Fluke within ± 0.4 °C (mean difference ≈ 0.1 °C). Two caveats apply: the per-node offset was tuned using the same Fluke, and the Fluke’s own accuracy (± 1.5 °C) exceeds the ≤ 1 °C target. This therefore demonstrates that the compensation reproduces the reference, but is not an independent accuracy validation; a rigorous ≤ 1 °C characterization over a wider range and across all three nodes would require a calibrated contact reference (thermocouple or reference RTD).

Use case. The continuous history captured a surface-temperature rise on a suction-fan motor, reaching a maximum of 70.0 °C (Fig. 5). The rising segments corresponded to periods when the machine stopped while the fan motor kept spinning unloaded; the pattern was later related to a bearing anomaly. This demonstrates the practical value of continuous logging over spot measurements.

Recovery and temperature range. After a node reset, the node boots, samples and transmits its first frame in about 2.279 s (measured from the firmware), well within one sampling cycle and far below any practical requirement. Across the three nodes, the recorded surface temperature spanned 15.3–70.0 °C, with per-node mean values of 39.5–45.4 °C.

Battery autonomy. A discharge test with a 1000 mAh cell (node in normal operation) showed the voltage dropping from 4.112 V to 4.089 V over 5 days (0.19 mV/h). Because the LiPo discharge is non-linear, autonomy was estimated two ways (Fig. 4): a linear extrapolation gives a conservative lower bound of ≈ 177 days, whereas modelling the typical LiPo voltage–state-of-charge curve (calibrated to the measured points) gives ≈ 290 –307 days (≈ 10 months). The implied average current is 0.13–0.24 mA, consistent with the measured 1.9% duty cycle (deep sleep ≈ 15 μ A for 98.1% of the time plus a 2.279 s active pulse). With the 2000 mAh battery of the prototype the autonomy would roughly double.

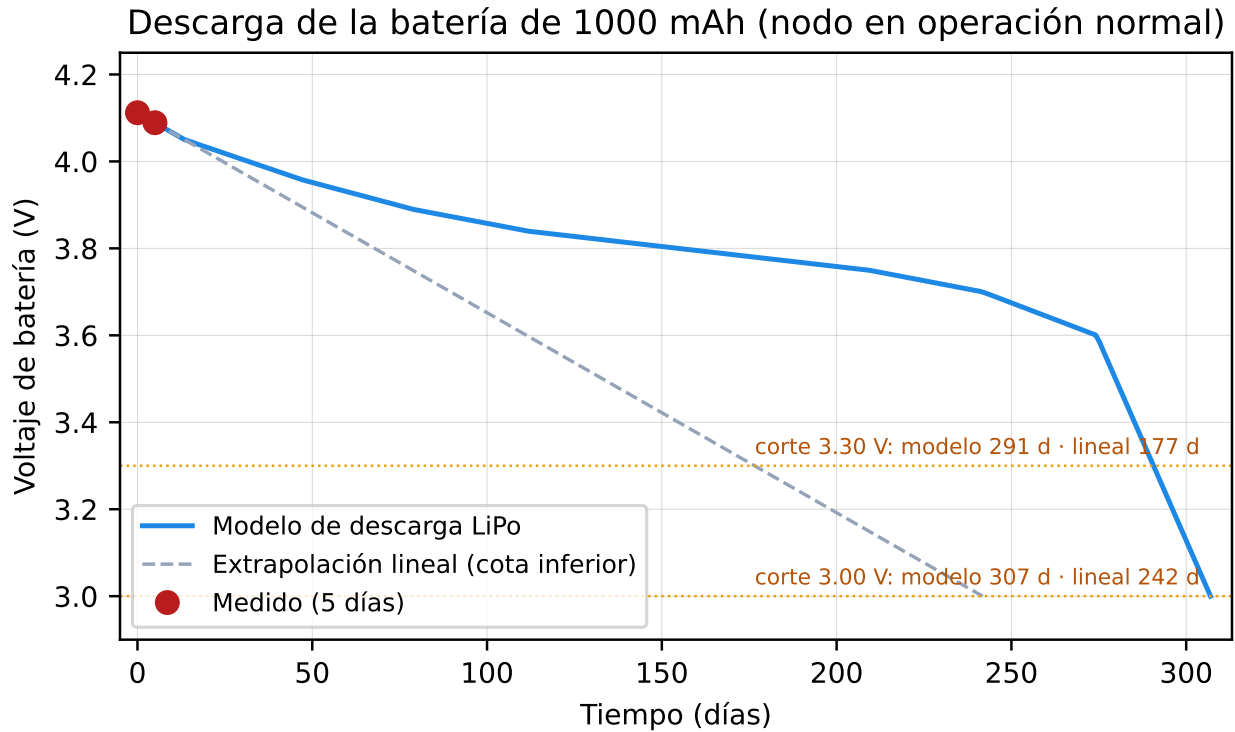


Figure 4: Battery discharge test (1000 mAh cell): measured points, non-linear LiPo discharge model (voltage–state-of-charge curve) and a linear extrapolation as a conservative lower bound. The model estimates an autonomy of about 10 months.

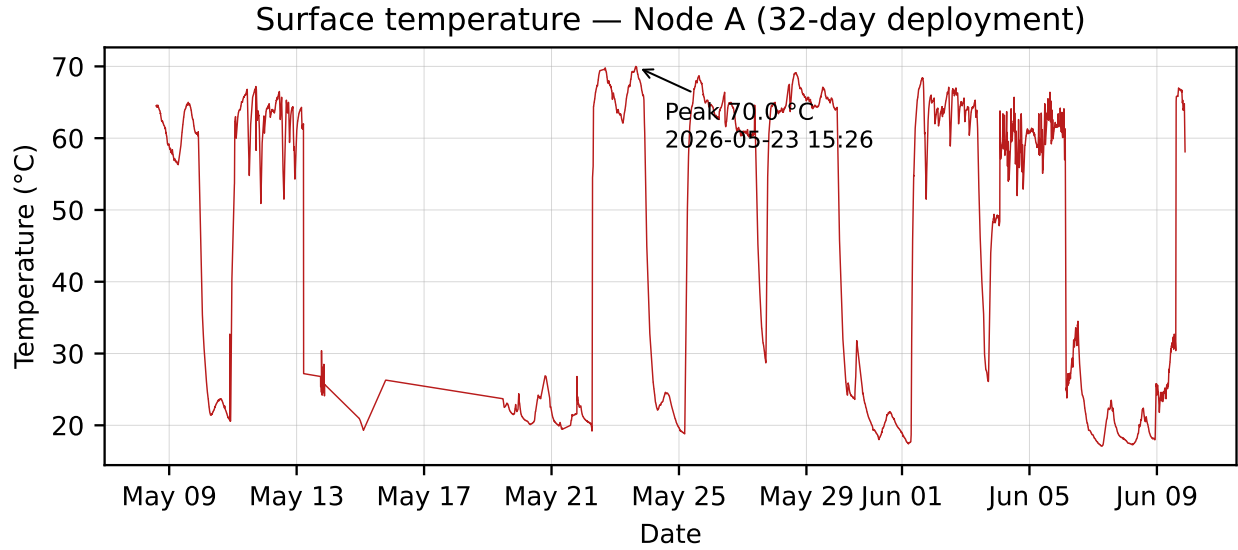


Figure 5: Surface temperature of one node over the 32-day deployment, showing the maximum associated with the suction-fan case.

Capabilities and limitations.

- Measures surface temperature with a three-wire PT100 over the MAX31865 range; per-node configurable offset.
- LoRa link availability $>97\%$ (active intervals) at a 121 s cadence over 32 days.
- Local operation with per-sensor daily history (up to 30 days on-device) and no Internet dependency.
- Temperature readings agreed with an infrared reference within $\pm 0.4\text{ }^{\circ}\text{C}$; a strict $\leq 1\text{ }^{\circ}\text{C}$ validation requires a calibrated contact reference (planned).
- Battery autonomy on the order of months (a 5-day discharge test gives ≈ 177 days conservatively and ≈ 10 months from the LiPo-curve model on a 1000 mAh cell; $\approx 0.13\text{--}0.24\text{ mA}$ average at a 1.9% duty cycle), far exceeding typical requirements.
- Limitations: absolute temperature accuracy depends on PT100 tolerance, mounting and per-node calibration; RSSI/SNR are not exposed by the current firmware; end-to-end latency was not measured (no node-gateway clock synchronization).

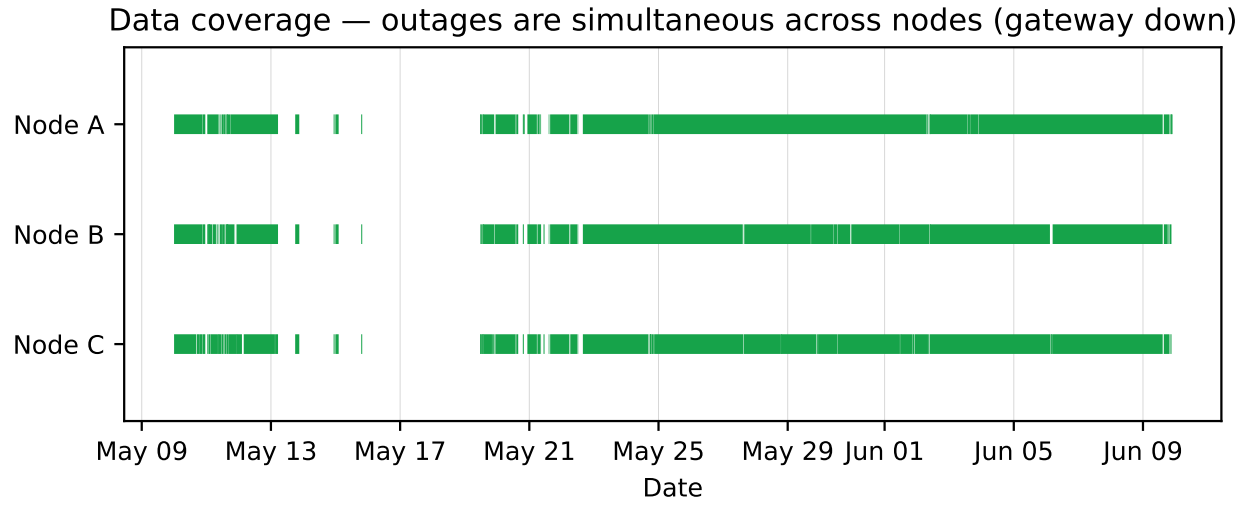


Figure 6: Data coverage of the three nodes. Outages are simultaneous across nodes, corresponding to gateway-off periods rather than LoRa link failures.

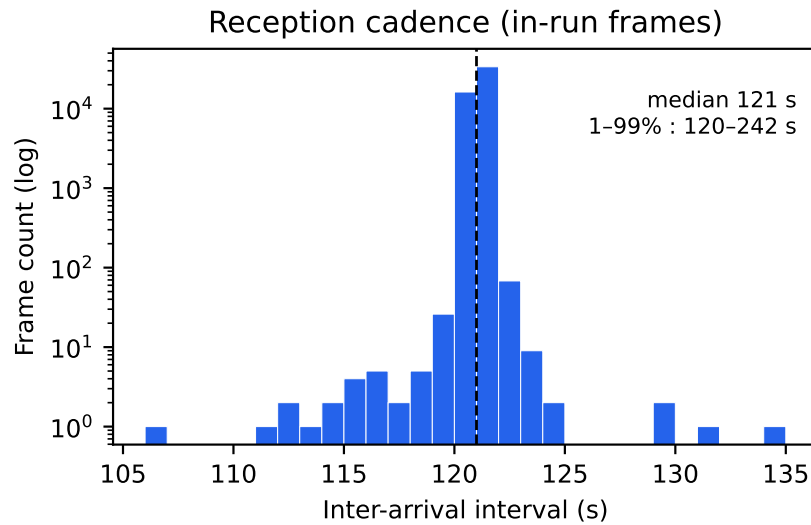


Figure 7: Inter-arrival interval distribution during active intervals (log scale); most frames arrive near the 121 s median.

Ethics statements

This work did not involve human subjects or animal experiments.

CRedit author statement

Jesus Andre Bojaca Lopez: Conceptualization, Methodology, Software, Investigation, Writing – Original draft. **David Santiago Puentes Cardenas:** Conceptualization, Hardware, Software, Validation, Data curation, Writing – Review & Editing. **Jhon Edwin Vera:** Supervision. **Sergio Francisco Mora Martinez:** Supervision.

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